HiCAP — Hierarchical Contrastive Action Priors on a frozen VLM backbone

A frozen Cosmos3-Edge reasoner read once per refresh, a 64 M trainable contrastive hierarchy selecting among 262,144 factored trajectories under prior-encoded pruning, trustworthy imagination and elastic sensing — plus the **v10 extensions**: agent supervision, an audio bridge, traffic-rule knowledge with a learned country label, optional reasoning text.

evidence class on every number: M measured · P published · E estimated · H hypothesis · I inherited

2.43 B frozen P · **64.37 M** trained M · **262,144** trajectories · **10 Hz** command · **2 Hz** refresh · **150** tests pass, 1 skipped (CPU) M

STATUS: architecture incl. v10 implemented and unit-tested on CPU; not yet trained

TanitAD programme · PI: Sayed · design & pre-registration poster · **2026-10-25**

v10 extensions implemented and integrated in `HiCAPFull` (spec rev b)

A · SENSE → ENCODE — the frozen backbone refreshes asynchronously; trainable tokens are rebuilt every tick

1 INPUTS

CAMERAS · 7 views, elastic
front-wide 120°, 256×640 → 160 tokens, always on
+ tele, cross L/R, rear L/R, rear-tele at 64 tok each
10 options T0–T3 · vision-only gate
cached today: front-wide only M

MICROPHONE · optional
mono or array · 2 s log-mel, 0.5 s hop · no audio in any corpus M

COUNTRY via GNSS · optional
supplied arm only; the country *predicted* from vision (s1) is primary; both arms reported

REFRESH CLOCK · 2 Hz · ASYNCHRONOUS, OFF THE COMMAND PATH

2 FROZEN Cosmos3-Edge reasoner 2 Hz
2.43 B P · prefill only, fixed short prompt, one frame per camera
SigLIP2 0.49 B → Nemotron-dense LM · 56 blocks
tap blocks → 28 · 42 · 56
LM 1.41 B body + 0.54 B embeddings · taps at blocks **28 / 42 / 56** → adaptive pool → **128 tokens × 3 depths** + last-token state
KV cache of the latest prefill kept K → explanations

3 TAP CACHE · 3 × 128 × 2,048 per refresh
reused between refreshes with an age embedding · int8

4 SHARED TEXT-EMBEDDING SPACE
$\mathbb{R}^{d_V}$, d_V = 2,048 · via the frozen text path, **cached once**
- **node definitions** · strategic + tactical (NodeLibrary)
- **country nodes × 25** · "driving in (country)"
- **rule embeddings r_i** · RuleBook v1, per country
- **audio bridge → 8 tokens** (trained bridge)
- **caption embeddings** · audio alignment targets

5 FROZEN audio encoder · optional
Qwen3-Omni AuT 647.9 M · BEATs ≈ 90 M contender

supplied arm: the country selects R_c

M

COMMAND CLOCK · 10 Hz · EVERY TICK (FAST PATH)

MOTION FRAMES · 3 × 64×160
front-wide, 100 ms apart

EGO · v, a_{lon}, yaw rate
no curvature (PI ruling 2026-10-25)

NAV COMMAND + 4 args
follow / left / right · stamped oracle

ROUTE CHECKPOINT RC-A50
x, y, ψ · σ 2 m / 0.75 m · drop ≥ 0.3 · oracle

Motion adapter Mφ · 0.6 M M 10 Hz
3 frames + 2 differences → 4 stride-2 convs → 4×4 pool → 16 tokens

16 motion tokens

Ego · nav · route tokens · 0.9 M M 10 Hz
Fourier features (8 octaves) → MLP, one token each
nav token = embedding(command) + MLP(d_{next}, d_{end}, $\Delta\psi_{next}$, valid)
route = MLP(x, y, ψ, valid), RC-A50 noised during training each has a **learned null row**: ego dropout, absent oracle, and the ego-zeroed, nav-zero and checkpoint-zero controls

ego · nav · route tokens (3)

6 TRAINABLE HiCAP FUSION TRANSFORMER Fθ 10 Hz

reads cached frozen tokens; no gradient reaches the backbone

1 last	128 VLM tokens softmax mix of 3 depths + age Fourier	16 motion	ego	nav	route	6 level queries s1 · s2lat · s2lon · s3P · s3V · goal

6 × block · LN → MHA → LN → FFN
d = 512 · 8 heads · FFN 2,048 · 18.9 M
cached-token projector 4.6 M M · fusion 18.9 M M
≈ 160 input tokens · ≈ 8 GFLOP per tick, launch-bound E

7 Gated memory read (zero-init)
$q'_\ell = q_\ell + \tanh(g_\ell) \cdot \text{CrossAttn}(q_\ell, M)$, $g_\ell = 0$ at init
M = **audio** 8 tokens (optional) ∪ **rule tokens** top-8 (v10) · per-level gates · gated rule memory 2.10 M M · source absent ⇒ output **bit-exactly** the original (unit test)
rule memory feeds s1 / s2 queries — never the goal token

Level readouts $s_\ell = \text{norm}(H_\ell(F_\theta(\cdot)[q_\ell]))$, H_ℓ: 512 → 1,024 → 512 · Matryoshka-nested widths 128 ⊂ 256 ⊂ 512 · five heads 5.3 M M

s1 strategic · 2 s	$s2^{lat}$ tactical · 0.5 s	$s2^{lon}$ tactical · 0.5 s	$s3^{P}$ path · 0.1 s	$s3^{V}$ profile · 0.1 s	goal 22 goals + point

Trainable parameters (`HiCAPFull`, counted from code) M: **56.39** core + **1.88** imagination & elastic + **6.10** v10 knowledge = **64.37 M** at d_{audio} 768

fusion 18.9	re-ranker 13.7	towers 9.2	Hℓ5.3	proj 4.6	v10 6.10

then: heads 1.9 · gated audio 1.4 · ego/nav/route 0.9 · motion 0.6 · **imagination + elastic 1.88** · **knowledge 6.10 = country readout 3.67 + rule memory 2.10 + retrieval 0.33**. With audio: GatedAudio widened to $d_{audio} = d_V$ = 2,048 (65.02 M) + AudioBridge 9.60 M ⇒ **74.62 M** M

← **Every tick**, 16 motion tokens and the ego, nav and route tokens join the sequence; the cached VLM tokens are re-read with their age, so the fusion never waits for a refresh. Audio and rules enter only through the gated read (7), never as input tokens: with the source absent the model is exactly the original. The parent level's choice is never a token.

6 unit-norm level readouts s1, s2lat, s2lon, s3P, s3V, goal → band B

B · DECIDE — coarse-to-fine retrieval → pruning → re-rank + imagination → verifier → command

8 s1 STRATEGIC every 2 s
7 declared / **5 usable** actions; band [8, 30] s, ≤ 12 s of future on this corpus; HOLD_MAIN_ROAD 2,468 / 4,719 clips I
node a_V = norm(G([t_V; φ(args)])), t_V from the shared space (4)
a parent restricts its children (hard beam), never fed as a feature

s2 TACTICAL · lat 3 × lon 7 every 0.5 s

	HOLD	CREEP	STOP	FOLL.	DEC.	ACC.	KEEP
LANE_KEEP							
TURN_L							
TURN_R							

21 cells, **19 populated**, all ≥ 30 train rows M · $f_2 = \tau(s2^{lat}, a_{lat}) + \tau(s2^{lon}, a_{lon}) + b(\text{lat, lon})$, b = −∞ for never-seen pairs

s3 TRAJECTORY · factored every 0.1 s
K_P = 1,024 paths × K_V = 256 speed profiles = 262,144, composed by arc-length re-parameterisation; the lateral class partitions paths, the longitudinal class partitions profiles
standstill sub-vocabulary hold / creep / stop (v_0 < 0.5 m/s) · keep (20, 10) → 1,480 rows, 177× fewer E

9 COUNTRY readout v10
on s1 · contrastive vs 25 country nodes · λ_c = 0.1 → p(c) · 3.67 M M

RULE RETRIEVAL v10
$w_i = \Sigma_c\, p(c) \cdot 1[i \in R_c]$
$s_i = \langle W_q\, q_{s2}, W_r\, r_i \rangle$
top-8 by w_i · softmax(s_i) → memory (7), prune, verify
projections 0.33 M M · RuleBook v1: 106 records

rules

goal query
22 goals (multi-label) + goal point (x, y, t) at 2 / 4 / 6 s — **predicted, never supplied**

R4

factored scores, classes → members

10 PRIOR-ENCODED PRUNING
no parameters · bitsets over a 1,024-bit path mask × 256-bit profile mask
- **R1 physics** (accel 6, brake 10, lateral 9 m/s²) + continuity tubes: removes 78 % of a speed-agnostic set, keeps **881 / 881** logged futures M
- **R2** state admissibility (conformal) · **R3 route** (nav-fed arm only)
- **RULES (v10)**: speed cap, right-on-red removal; only removes what R1 kept
- **R4** geometry goals (soft) · **R5** hysteresis (soft)
- **R6 conformal keep-sets**, Mondrian, episode-blocked: recall ≥ 1 − $\Sigma\alpha_\ell$
- **GUARD SET always survives**: hold-CTRA, gentle / firm / max brake, previous plan

262,144 → ≈ 200 → ≤ 32 → K = 16 (32 when the margin is small)

≈200

11 RE-RANK · MaxSim shortlist ≤ 32 → cross-attention
candidates (path, profile, cell embeddings + 8 slots) attend to scene tokens · 3 layers · MoE FFN routed **deterministically by lateral class** (3 experts) · 13.7 M M, stage 3 only
sub-scores ttc · clearance · progress deficit + logit · train K = 32, deploy K = 16 · 0.34 GMAC per tick E

12 IMAGINATION · three tiers, cheapest first

T0 train only	outcome-aware soft target $y(a) = 0.7 \cdot \text{softmax}(-J_{teacher}/T) + 0.3 \cdot 1[a = a^*]$ · zero inference cost
T1 every tick	E-head 512 → 1,024 → 24 on the state only: lead gap, closing speed, lead accel at 0.5 / 1 / 2 / 3 s, mode, onset q10/50/90 → $J_1 = E_{m,\tau}[c(e_a, x_{m,\tau})]$, analytic
T2 gated, top-8	two counterfactual cost heads; disagreement = **VETO**, never a trigger; learned benefit gate ≤ 30 % of ticks

score = τ(u, v) + log priors − $\beta_1 J_1$ − $\beta_2 J_2$ → feasibility mask **last**

if

13 SYSTEM-2 VERIFIER
fires **only if** top-2 margin < m_{min} or the T2 heads disagree
top K_2 ≤ 8 survivors rendered as text ("lat=LANE_KEEP, lon=..., target 6 m/s") + retrieved rules, after the cached scene prefix
frozen VLM adds λ · match score: **scores, never generates**
≈ 58.6 ms (8 B, bf16) E · deadline-bounded

certain ticks: no verifier

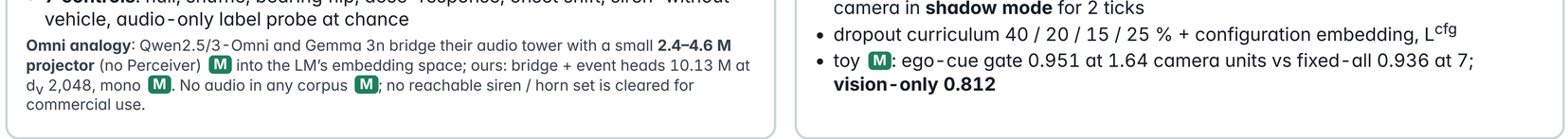
14 COMMAND
6 s trajectory @ 10 Hz (60 steps; contract slots 0.5 … 6 s)
$\hat{y} = \pi_{prior}(e_t) + \rho(v_3) + \delta$: CTRA prior + leaf residual + refinement

selected nodes
strategic node · tactical cell (lat, lon) · path p, profile q

confidence
from the base score only (guard G5)

explanation text K
≤ 48 tokens, ≤ 0.5 Hz, from the refresh KV cache · **off the control path**, one-way

TWO CLOCKS command latency L_{cmd} does not depend on backbone size to first order; the backbone only sets token age A and duty D · illustrative: Edge, 3 cameras, T_r ≈ 62 ms, o = 15 ms E



COLOUR KEY
- FROZEN 🔒 reasoner, audio encoder, verifier
- CACHED ONCE text embeddings, Stage-0 tables
- ANALYTIC / RULES no parameters
- TRAINED fusion, heads, critics, gates
- OPTIONAL mic, GNSS, verifier, explanation
- 10 Hz path · 2 Hz refresh · dotted = optional

a Multi-hierarchy and contrastive critics (8 11)

Every level is a **dot-product critic** between a unit-norm state readout and cached node embeddings, trained with **level-wise negatives**: a sibling-only softmax leaves a per-parent offset free and mis-orders nodes across parents.

$$\tilde{p}_\ell(v \mid s) = \text{softmax}_{v' \in V_\ell}(\tau_\ell(s_\ell, a_{v'}) + \beta_\ell \log \tilde{q}_\ell(v')),\ \tau = e^{\theta}, \text{ init } 1/0.07, \text{ clamp } 100$$

- **Soft targets**: $y(a) \propto \exp(-d_{tw}(y^*, \hat{y}_a)^2 / 2\sigma^2)$, σ = 0.5 m, summed up to ancestors ⇒ β = 0 by default
- **Consistency**: Σ_ℓ KL(sg[$\hat{p}^{cascade}_\ell$] ‖ $\tilde{p}_\ell$) ties critics to the beam
- **Goal loss**: 22 goals, BCE (positive weight ≤ 50) + goal point at 2 / 4 / 6 s
- **Matryoshka** readouts at 128 / 256 / 512: a free option, expected to save nothing
- Masked vs unmasked InfoNCE vs soft-target softmax is an **ablation (H-HC4)**, not assumed

$$L = \Sigma_\ell \lambda_\ell L^{voc}_\ell + \lambda_c L^{cons} + \lambda_g L^{goal} + \lambda_k L^{cost} + \lambda_r L^{ref} + \lambda_i L^{img} + \lambda_a L^{aud} + \lambda_f L^{cfg}$$

$$a^* = \text{argmin}_{a \in K} \Sigma_k w_k \hat{c}_k(s, a) - \lambda \log \hat{p}(a \mid s),\ \ w \text{ fixed a priori, } \lambda \text{ fit on held-out } \textit{train} \text{ episodes}$$

b Knowledge injection: three routes (4 7 9 10 13)

text-defined nodes · rules · countries
- ① **MEMORY** gated cross-attn into s1/s2 queries (7)
- ② **PRUNE** numeric args → masks / priors (10)
- ③ **VERIFY / EXPLAIN** rule texts in prompts (13, K)

- **NodeLibrary**, versioned: ids never move
- **RuleBook v1**: **106 records, 25 countries** — 73 PUBLISHED (official-domain search summaries), 33 UNVERIFIED M; {rule_id, countries, text, args, road_type, effect ∈ {prune, prior, memory}, source}
- **Country**: 25-way readout on s1 (λ_c 0.1); labels for 4,508 / 4,508 clips on private HF …/`tanitad-refcv8-v10-labels` I
- **Country-swap counterfactual (H-HC20)**: swapping p(c) may change the decision *only* on rule-relevant cells

Content: driving side · default speed limits · right-turn-on-red · priority to the right · roundabout yield · emergency vehicles · pedestrian crossings · tram priority · school bus (US) · daytime lights.

Corpus limits H: all 25 countries drive on the right; right-on-red differs only for 265 US clips (no traffic-light labels); speed limits are testable.

c Imagination that can be trusted (12)

Only the **exogenous** future is learned; the ego path is a known function of the candidate. Toy: **most of the gain is the target shape**.

tier	toy result (3 seeds) M
T0 soft targets	regret 7.00 → 3.43 (paired Δ 3.57, 95 % [2.99, 4.19])
T1 E-head + J_1	regret 1.80 vs 47–72 for unstructured arms
T2 veto	AUROC 0.88 (recall 64 % at 5 % FP); learned gate: 65 % of gain at 30 % budget

- **G1** feasibility mask last: infeasible over-braking 89.8 % → 0 % of toy ticks
- **G5** confidence from the base score only: an imagined cost flipped corr(conf, regret) −0.11 → +0.19
- **No gradient** from imagination into the selector or backbone
- **v10 supervision**, eval139 M: lead agrees with v9 on 90.1 %; gap within 2 m on 92.1 % after a constant +3.55 m offset; logged trajectory collides on 0.26 %; modes = MODE_ACC (brake / hold / accelerate / brake hard), cut-outs IGNORE; agent files for 4,508 / 4,508 clips; J open-loop, non-reactive

Also G2 coverage audit · G3 shuffled cue · G4 veto ≠ trigger · G6 β on *train* · G7 marginalise all · G8, G9.

d Audio bridge into the shared space (5 4 7)

default (a): HiCAP fusion read · ablation (b): 8 soft tokens appended to the VLM prefill (omni pattern)

mic 2 s / 0.5 s → FROZEN encoder log-mel → features → TRAINED bridge perceiver + MLP, 9.60 M → 8 tokens, width d_V → gated read, g = 0

tower: Qwen3-Omni AuT 647.9 M, Apache-2.0 (recommended) · BEATs ≈ 90 M · E-N0 decides

- **A1** align: symmetric InfoNCE + cosine to the frozen text path's caption embedding (AudioCaps, Clotho, AudioSet, ...)
- **A2** event heads (class, bearing, distance, approaching) from event metadata only
- **A3** gate on synthetic events rendered from **real agent tracks**, supervised through knowledge ("yield to emergency vehicles"), never a trajectory
- **7 controls**: null, shuffle, bearing flip, dose-response, onset shift, siren-without-vehicle, audio-only label probe at chance

Omni analogy: Qwen2.5/3-Omni and Gemma 3n bridge their audio tower with a small **2.4–4.6 M projector** (no Perceiver) M into the LM's embedding space; ours: bridge + event heads 10.13 M at d_V 2,048, mono M. No audio in any corpus M; no reachable siren / horn set is cleared for commercial use.

e Elastic sensing: cameras as a decision (1)



10 options, 160 + 64·(n − 1) tokens. A camera gate maps a situation to a camera set — a **situation classifier in disguise** — so the primary gate **G-V** reads only the front-wide embedding and its own previous option.

- loss $\Sigma_j p_j[E_j(x) \cdot \text{crit}(x) + \lambda\, \text{cost}_j]$ on cached per-option errors
- **keep-alive**: escalate in 1 tick, de-escalate after 3 ticks with margin 0.10; new camera in **shadow mode** for 2 ticks
- dropout curriculum 40 / 20 / 15 / 25 % + configuration embedding, L^{cfg}
- toy M: ego-cue gate 0.951 at 1.64 camera units vs fixed-all 0.936 at 7; **vision-only 0.812**

h Latency and budget: two clocks (2 6 13)

A = T_r + o + 1/f_r · D = $f_r(T_r + o)$ ≤ 0.5 · L_{cmd} p95 ≤ 100 ms · fast path p95 ≤ 25 ms *while a refresh runs*

Cosmos3-Edge reasoner, ViT + prefill	1 cam	3 cam	3 cam × 3 fr
FP16 (NVIDIA CI precision) E	41	76	180 ms
NVFP4 (untested) E	22	43	109 ms

duty in one 500 ms period at 2 Hz (o = 15 ms):

Edge, 3 cam: (62 + 15) ms → D ≈ 0.15 · limit 0.5 · 8 B, 3 cam × 3 fr: D = 0.48

- 8 B-class at 3 cameras: A ≤ 90 + 15 + 500 ≈ 0.6 s ≈ 9 m at 15 m/s E
- fast path: fusion ≈ 8 GFLOP · cascade 0.34 GMAC · imagination ≈ 38 MFLOP E
- verifier ≈ 58.6 ms, outside the L_{cmd} claim; a deadline miss holds the last command

Rows scale NVIDIA-published Thor figures; **nothing here is a HiCAP timing**. The on-device bench (H-HC11) is written, not run. Edge bracket 28 / 62 / 130 ms at 1 / 3 / 7 cameras E; D ≈ 0.15 is arithmetic on it.

f Training schedule

losses.STAGES

component / loss	S0	A1	1	2	3	opt
tap caches, text embeddings, vocabulary, imagination targets, country labels	■					
audio bridge alignment (backbone + HiCAP frozen)		●				
critics on all levels, goal loss, country readout (λ 0.1); Tier-0 + camera gate when supervised			●	●	●	
hard negatives, consistency KL, 40 % replay				●	●	
cost, refinement, L^{cfg}, audio, E-head, Tier 2; re-ranker only here					●	
rule-memory + audio gates (start at 0)				once attached		
student distillation; text-channel LoRA for explanations						◌
Cosmos3-Edge reasoner, audio encoder	■	■	■	■	■	■

Stage 0 tap caches: 148 GB int8 at 2 Hz, 188,640 refreshes E, once per backbone and camera set.

■ built once · ● trained · ■ frozen. AdamW, no weight decay, lr = $2 \cdot 10^{-3}$ √(1024/width) √(batch/1024), one-cycle, batch 2,048, 3 seeds + 1 replicate. Schedule = code (`losses.STAGES`, spec rev b §6).

g Data: v9 labels, v10 by reference

spec §0, §1, §4

v9 I: 4,369 train / 139 eval clips, 869,278 rows at 10 Hz (≈ 24.1 h). Census M: lane keep 632,971 · turn L 48,830 · turn R 47,726.

v10 M: **country** 25 countries, 4,369 / 4,369 + 139 / 139 joined (100 %) · **agent cuboids** 4,585 / 4,719 clips, 29.4 M rows · **Alpamayo CoC** 4,729 clips × 1 anchor · **audio: none**.

trained part	missing	filled by
core HiCAP	**VLM tap caches**	pod job E27 — blocks all training
imagination T0–T2	**full train build**	producer written; eval139 + 100-clip sample built
elastic gate	**6 other cameras**	upstream chunks via camera_ref
node towers, rules	**embedding job**	GPU minutes; RuleBook v1 written
audio bridge	**all of it**	external sets + synthetic events
country · reasoning	**nothing**	labels published; zero-shot explanations

Explanation teacher (Alpamayo2-Super meta-action) vs v9 M: longitudinal 0.427 (κ 0.255, n 4,363), lateral 0.855 (κ 0.314, n 3,803). Per-frame regeneration pipeline built, **not launched**: ≈ 374 H100-h at 10 Hz; auto_labeling eligible on ≈ 61.6 % of rows, meta_action ≈ 93.3 % E.

i Admissibility rules

binding

1. The **goal / route path never reads** the situation or country posterior; there is no situation-classifier head; rule memory never feeds the goal token.
2. **Vision-only gate**: G-V reads frames only; G-E (ego) is an ablation pending a ruling; G-T is not provided.
3. **Audio-off primary numbers**; audio-on is a paired arm; the audio-only label probe must sit at chance.
4. Labels may use ego future and agents; inference sees frames, v, a_{lon}, yaw rate and the **stamped oracle** nav / route, each with zeroed and shuffled controls.
5. A parent restricts its children but is **never a feature** of the child readout; explanations never feed control.
6. The **country label** (upstream metadata) is never an inference input except in the supplied GNSS arm; both arms reported.
7. Four metric families, **per family, never pooled**; paired episode-cluster bootstrap on the full-set mean.

j Pre-registered (both outcomes committed) — and what is NOT yet measured

sec_protocol, spec §3.4

H-HC0 gate: frozen + motion + v_o k-NN vs v_o-only
H-HC1 motion adapter, ego zeroed
H-HC2 driving knowledge (Alpamayo pair)
H-HC3 tree vs flat on 262,144 entries
H-HC4 prior correction (InfoNCE variants)
H-HC5 vocabulary scale 2^{10} ... 2^{18}
H-HC6 pruning retention + coverage
H-HC7 re-score vs dot product
H-HC8 imagination (E-I0 probe first)
H-HC9 student (conditional)
H-HC10 elastic sensing (oracle camera first)
H-HC11 latency, contention, precision on Thor
H-HC12 audio T1–T4 + controls
H-HC13 data efficiency 3 / 10 / 30 / 100 %
H-HC14 closed loop, n ≥ 30 scenes
H-HC15 token age 0 / 0.5 / 1.0 / 1.5 s
H-HC16 backbone scale ladder 2 / 4 / 8 B
H-HC17 partial adaptation (LoRA)
H-HC18 country acc. (vision-only) vs 1/25
H-HC19 rules on / off, paired per family
H-HC20 country-swap counterfactual
H-HC21 explanation channel

NOT MEASURED no HiCAP model trained · no accuracy or driving metric · no Thor timing · tap caches never built · E-I0 not run · dense reasoning not launched

Sources: Paper/HiCAP/hicap.tex (sec_method, sec_imag, sec_elastic, sec_eff, sec_problem, sec_protocol, sec_limits) · HICAP_V10_SPEC.md rev b · stack/tanitad/hicap/ · DataEng/v10/ · RG note

TanitAD · HiCAP poster · 2026-10-25 · nothing trained; every result shown is a toy, a mask check, a census or arithmetic